Culture Cartography: Mapping the Landscape of Cultural Knowledge

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How should we build **cultural knowledge banks**?

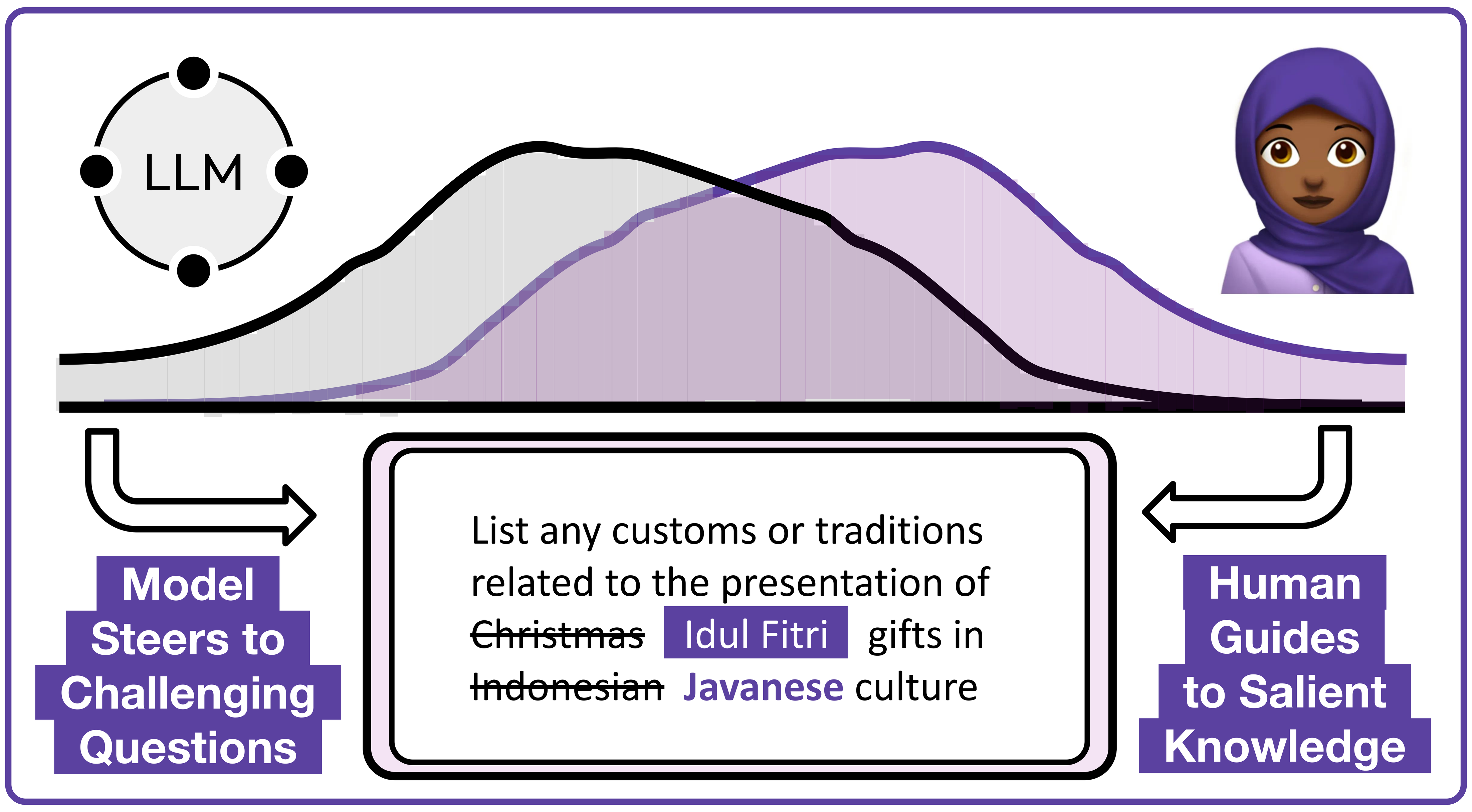
{ Mixed-Initiative Data Elicitation }

(Jointly Produce Knowledge)

{ Single-Initiative }

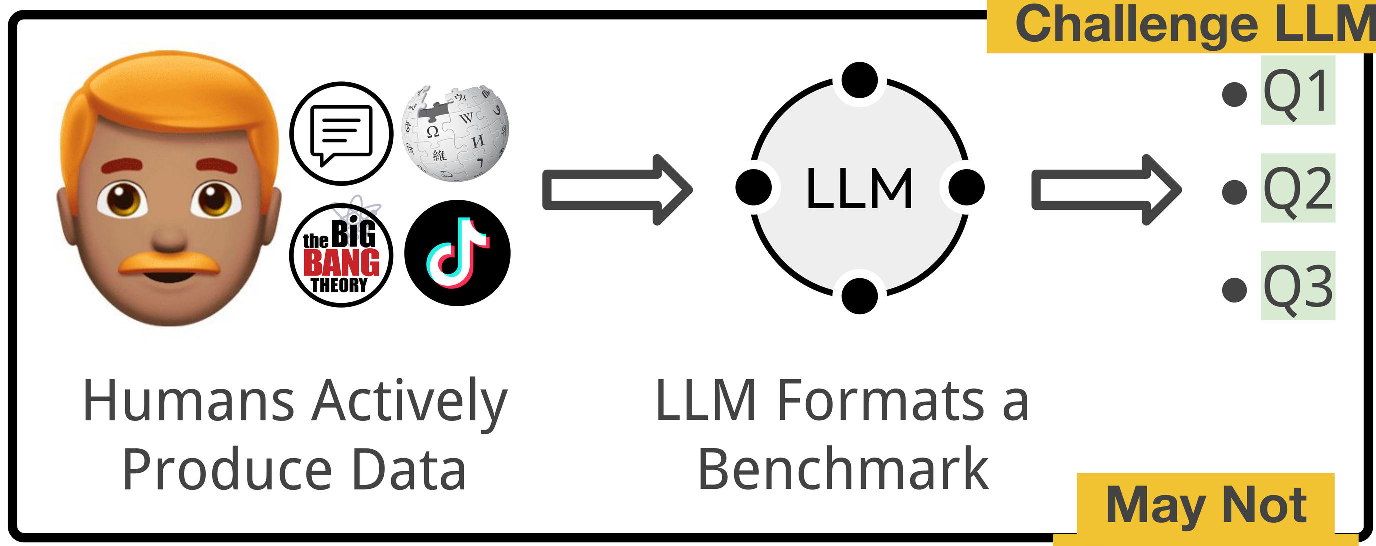
(Human or LLM Produces)

Culture Cartography



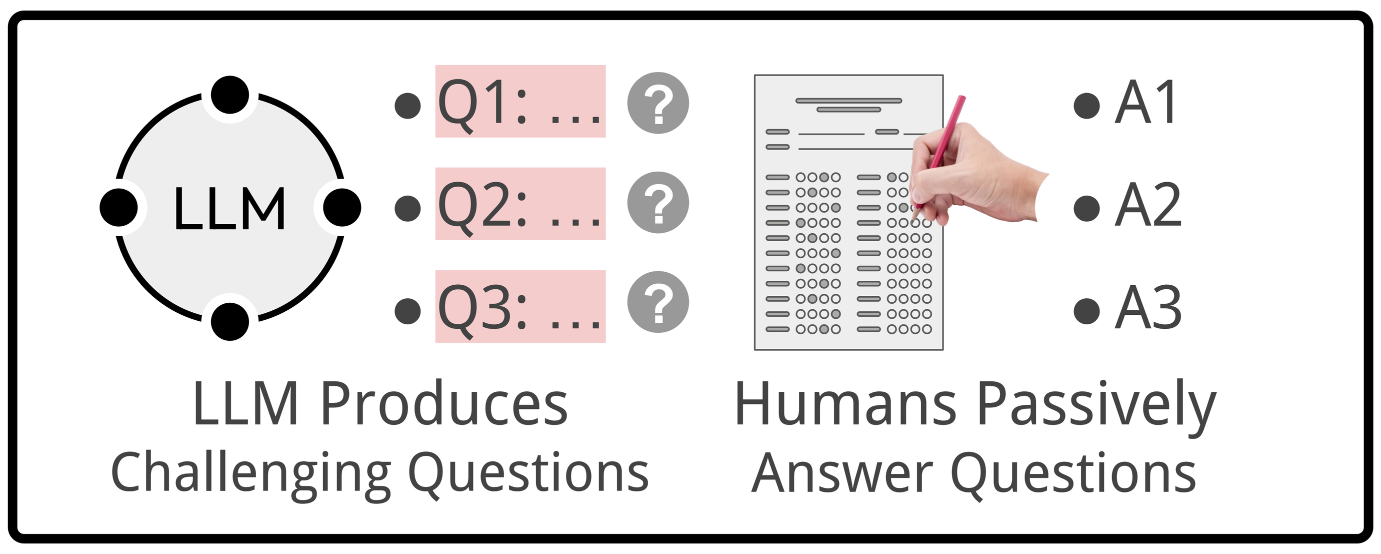
Knowledge Extraction

May Not Challenge LLMs



May Not Represent Culture

Traditional Annotation



Finding 1: Culture Cartography produces more *long-tail* data than *Traditional Annotation.*

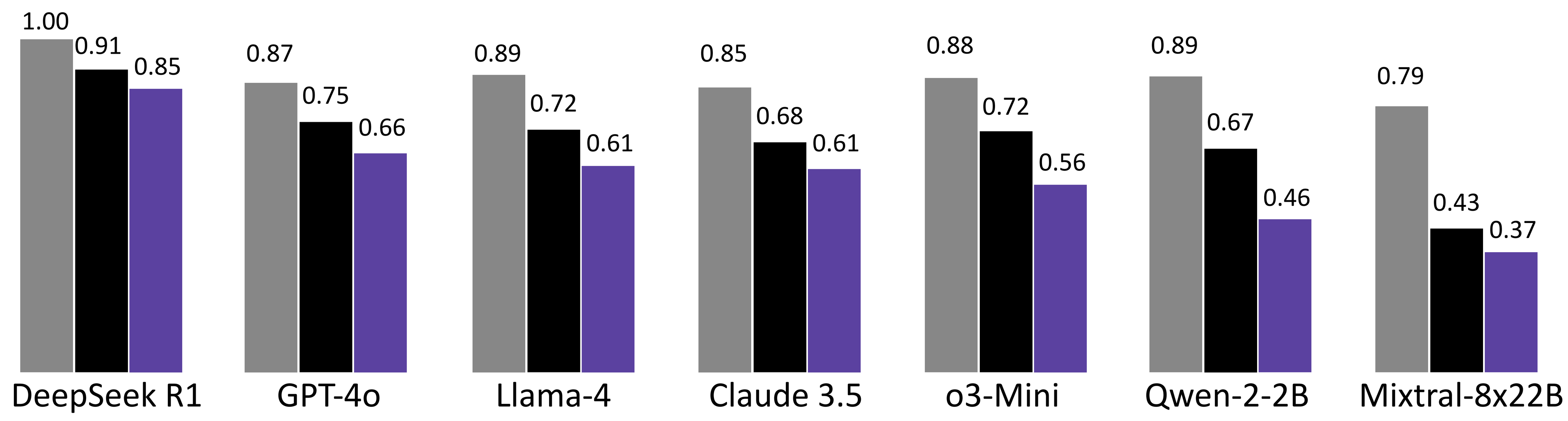
Evaluation: *Recall@100*

For every cultural question in the data subset, the model generates 100 candidate answers. Against the ground truth answers in the subset, we compute the model's recall: how many gold answers did it generate?

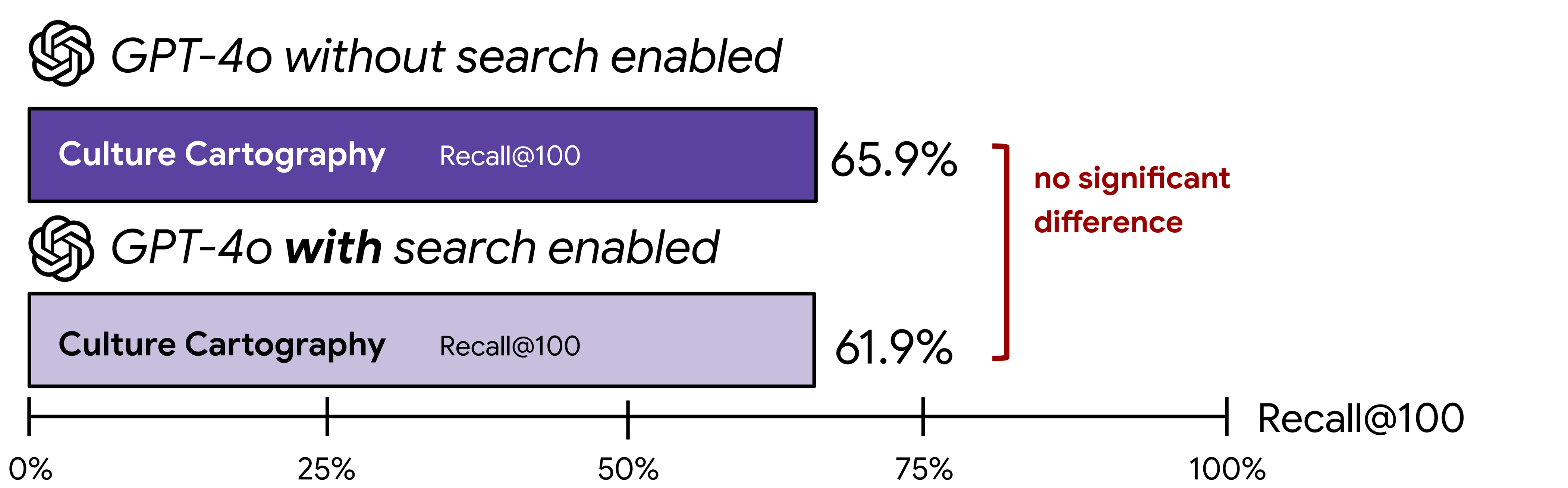
LLM-Generated Data: Humans validate the top-four LLM answers to a predetermined culture question about behavioral norms.

Traditional Annotation: For each LLM-answered question, humans provide up to four new answers that complement what the AI already gave.

Culture Cartography: Annotation with our system. (1) The LLM proposes low-confidence questions; (2) the human makes direct edits or proposes new questions that reflect their expertise; (3) these edits constrain subsequent LLM generations.



Finding 2: Culture Cartography data is *Google-proof*, unlike Knowledge Extraction.



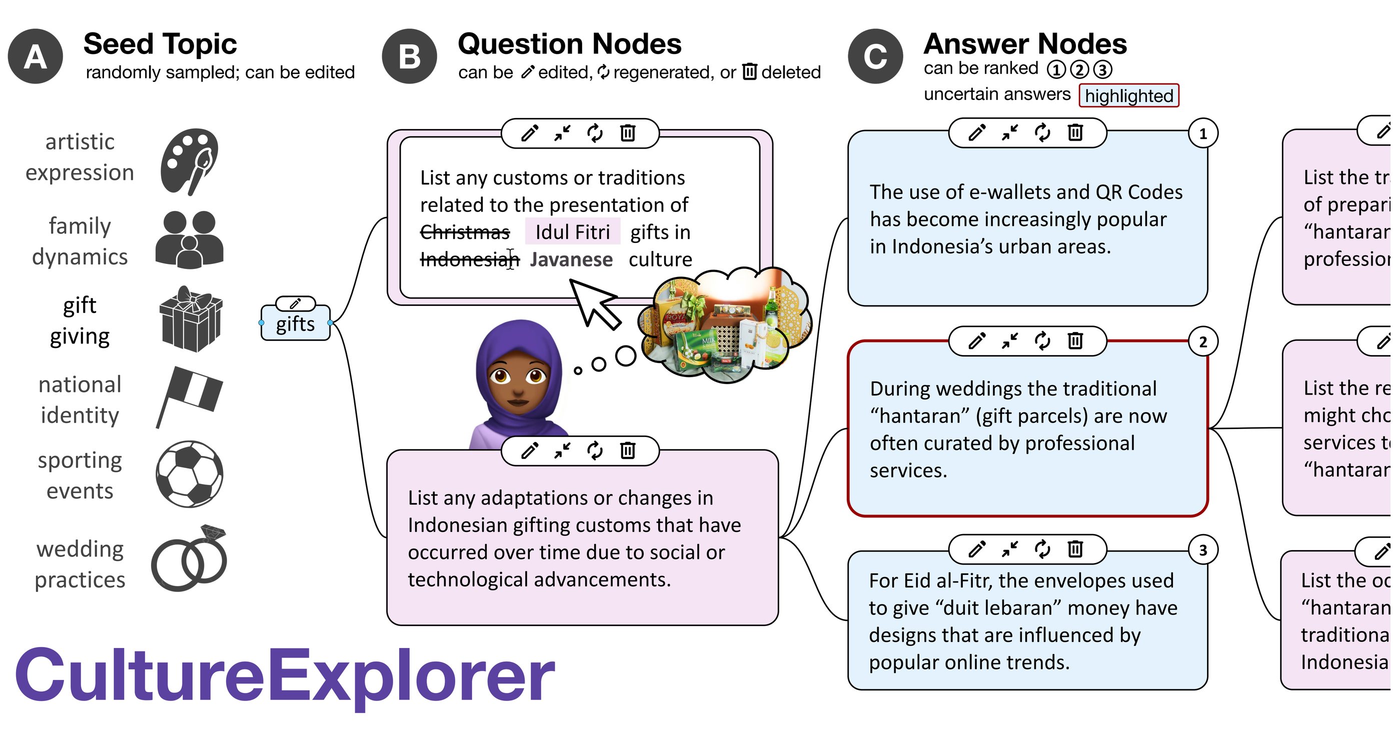
Finding 3: Cartography is aligned with prior work.

Evaluation: *Transfer Learning*

We show that supervised fine-tuning followed by DPO on Culture Cartography can boost the downstream performance of Llama-3.1-8B on two related culture benchmarks, BLEnD (Myung et al., 2024) and CulturalBench (Chiu et al., 2024). Here we show results on the Nigerian (🇳🇬) and Indonesian (🇮🇩) subsets, which correspond to cultures represented in our collected data.

	BLEnD		CulturalBench	
	🇳🇬	🇮🇩	🇳🇬	🇮🇩
Vanilla Model	56.0%	66.5%	50.0%	46.2%
Traditional Annotation	59.3%	**73.2%***	50.0%	**65.4%***
Culture Cartography	**62.5%***	**73.6%***	**63.6%**	**65.4%***

Try our web tool!



CultureExplorer

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